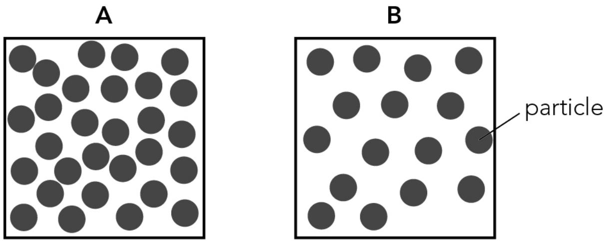
Name ______________________________________ Date ______________

Worksheet 1A: Calculating density

1 The figure shows two objects of the **same** type with the **same** volume:



Fill in the gaps below using these words.

particles **sink** **density** **float** **more** **less** **A** **B**

You can use each word once, more than once or not at all.

_______________ is mass per unit volume.

This tells us how close the _______________ are packed together.

Object A has _______________ particles, and therefore _______________ mass.

Object B has _______________ particles, and therefore _______________ mass.

So object _______________ is less dense than object _______________.

If the object is less dense than a liquid it will _______________.

If the object is more dense that a liquid it will _______________.

2 Predict if these objects float or sink in water and explain why you think this.

Density of water = 1000 kg/m^3

a Ice: 881 kg/m^3

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b Chalk: 1121 kg/m^3

____

c Wax: 320 kg/m^3

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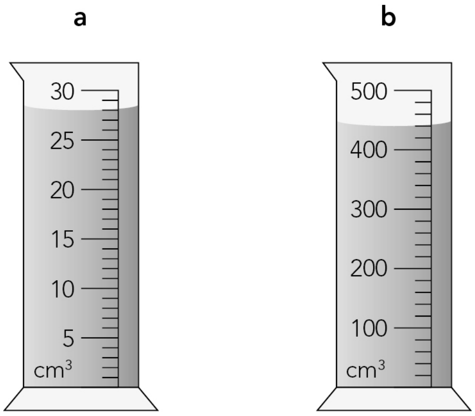
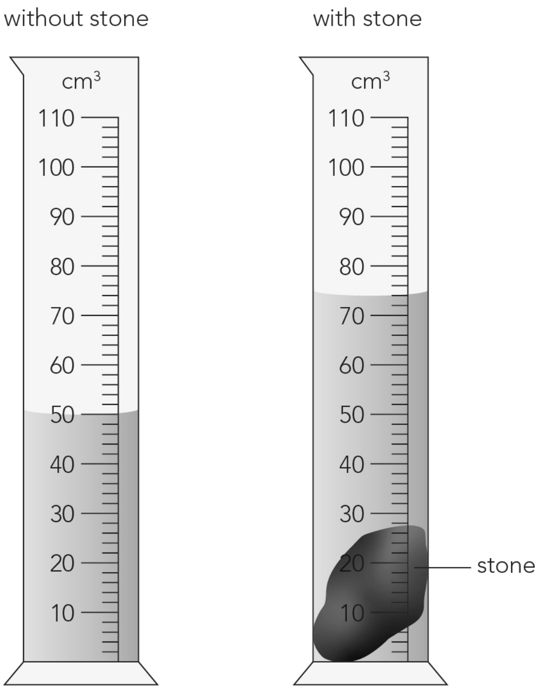
3 What are the volumes of the liquids inside these measuring cylinders?

Write your answers in the spaces below each cylinder.



_______________ _______________

4 The figure shows how the displacement method can be used to find the volume of a stone.



Calculate the volume of the stone in cm^3.

5 Use the equation for density to answer these questions.

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

a A tennis ball has mass 60 g and volume 160 cm^3. Calculate the density of the tennis ball. Give the units.

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b An ice cube has dimensions 3 cm × 3 cm × 2 cm. The mass of the ice cube is 16.2 g. Calculate the density of ice.

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